

L06 DAYLIGHT SIMULATION | O (MAX : 2 PT)

Intent : Ensure indoor daylight exposure through daylight simulation strategies.

Summary : This WELL feature requires projects to conduct daylight simulation calculations to make informed decisions around fenestration and shading, so as to provide appropriate daylight exposure for occupants.

Issue : The way buildings are designed has an enormous impact on occupant exposure to daylight.¹ Exposure to daylight has been proven to have a substantial impact on mood, circadian health and productivity.^{1,2} Studies have associated lack of exposure to daylight with a disruption in the circadian rhythms of humans and a decrease in sleep quality. Students have been shown to perform better when they have access to windows that can be opened.³ Indoor spaces with daylight exposure have been shown to have fewer bacteria, compared to spaces without any light exposure.⁴

Solutions : Building design and planning has a substantial impact on the amount of daylight in an indoor space. With the myriad of solutions available for daylight ingress and shading, indoor daylight exposure can be optimized by conducting simulations. Daylight simulations can inform choices for shading, support fenestration design, as well as the interior layout of a spaces.⁵⁻⁷

PART 1 CONDUCT DAYLIGHT SIMULATION (MAX : 2 PT)

For All Spaces except Dwelling Units & Guest Rooms:

The project demonstrates, through computer simulations, that the following conditions are achieved:

a.

The total floor area of regularly occupied spaces achieves one of the following targets:

Tier	Calculations per IES LM-83	Calculations per Annex A of CEN 17037	Points
1	sDA _{300,50%} > 55%	OR Target illuminance of 300 lux is achieved for >50% of regularly occupied areas throughout 50% of daylit hours of the year	1
2	sDA _{300,50%} > 75%	OR Target illuminance of 300 lux is achieved for >50% of total area and average illuminance 100 lux is achieved for >95% of total area throughout 50% of daylit hours of the year	2

For Dwelling Units & Guest Rooms:

The project demonstrates, through computer simulations, that the following conditions are achieved:

a. The total floor area of regularly occupied spaces of each dwelling unit achieves one of the following targets:

Tier	Calculations per IES LM-83	Calculations per Annex A of CEN 17037	Points
1	sDA _{300,50%} > 55%	OR Target illuminance of 300 lux is achieved for >50% of individual unit area throughout 50% of daylit hours of the year	1
2	sDA _{300,50%} > 75%	OR Target illuminance of 300 lux for >50% of individual unit area and average illuminance 100 lux is achieved for >95% of individual unit area throughout 50% of daylit hours of the year	2

REFERENCES

1. Boubekri M, Cheung IN, Reid KJ, Wang CH, Zee PC. Impact of windows and daylight exposure on overall health and sleep quality of office workers: A case-control pilot study. J Clin Sleep Med. 2014;10(6):603-611. doi:10.5664/jcsm.3780
2. Amundadottir ML, Rockcastle S, Khanie MS, Andersen M. A human-centric approach to assess daylight in buildings for non - visual health potential, visual interest and gaze behavior. Build Environ. 2016;113:1-40.
3. Heschong L. Daylighting in Schools: An Investigation into the Relationship between Daylighting and Human Performance. Detailed Report. Hmg-R-9803. 1999:140. <http://eric.ed.gov/?id=ED444337>.
4. Fahimipour AK, Hartmann EM, Siemens A, et al. Daylight exposure modulates bacterial communities associated with household dust 06 Biological Sciences 0605 Microbiology. Microbiome. 2018;6(1). doi:10.1186/s40168-018-0559-4
5. Sherif A, Sabry H, Wagdy A, Mashaly I, Arafa R. Shaping the slats of hospital patient room window blinds for daylighting and external view under desert clear skies. Sol Energy. 2016;133:1-13. doi:10.1016/j.solener.2016.03.053
6. Wagdy A, Sherif A, Sabry H, Arafa R, Mashaly I. Daylighting simulation for the configuration of external sun-

breakers on south oriented windows of hospital patient rooms under a clear desert sky. *Sol Energy*. 2017;149:164-175. doi:10.1016/j.solener.2017.04.009

7. Shen E, Hu J, Patel M. Energy and visual comfort analysis of lighting and daylight control strategies. *Build Environ*. 2014;78:155-170. doi:10.1016/j.buildenv.2014.04.028